Impact of Iron Availability on the Motility of Uropathogenic *Escherichia coli*

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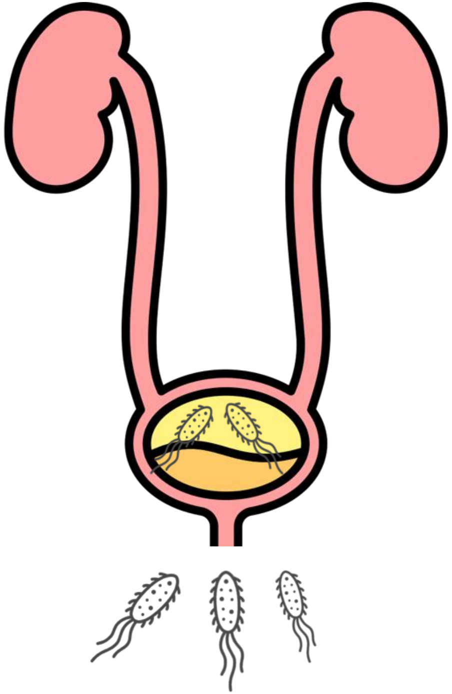
Cal Poly

Urinary Tract Infections



Uropathogenic *E. coli*

Low [Fe]

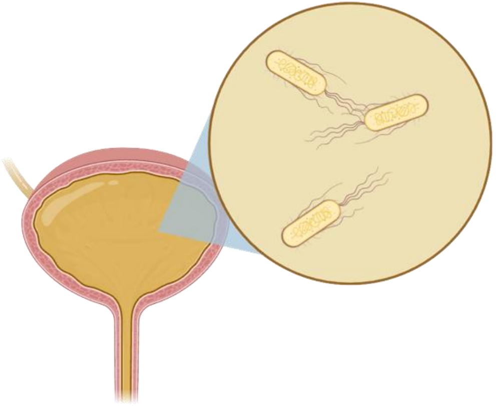
[Fe]

UTI therapy

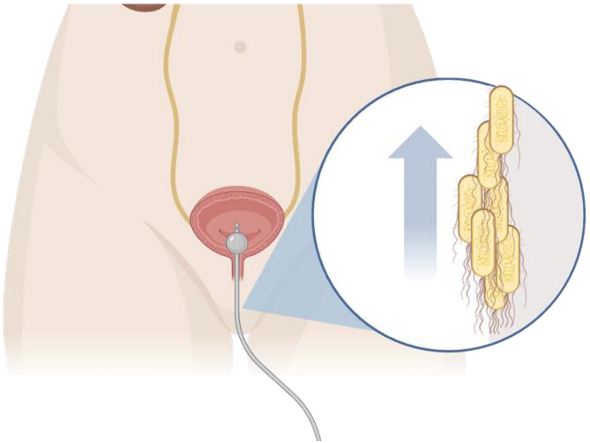
See posters 113 and 162 for more details

Motility and attachment are virulence factors for UTIs

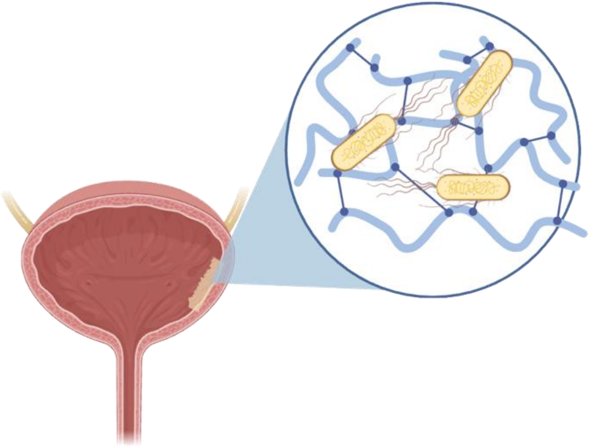
Swimming



Swarming



Attachment

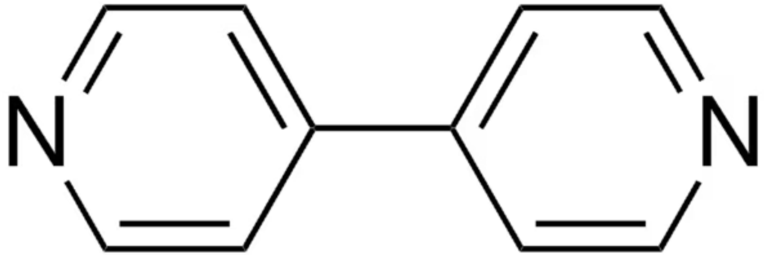


We are investigating how iron conditions affect these virulence factors

Iron (III) Sulfate

$$Fe_2(SO_4)_3$$

Dipyridyl (iron chelator)



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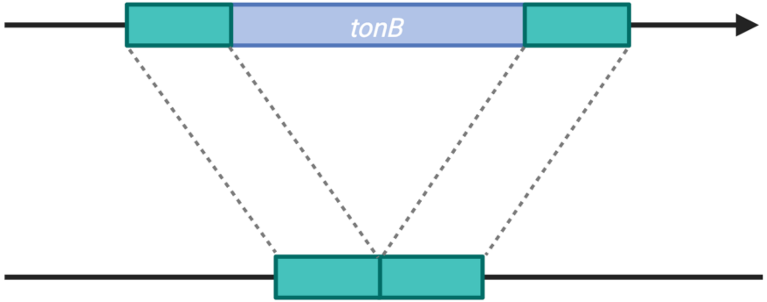
Meet the mutants

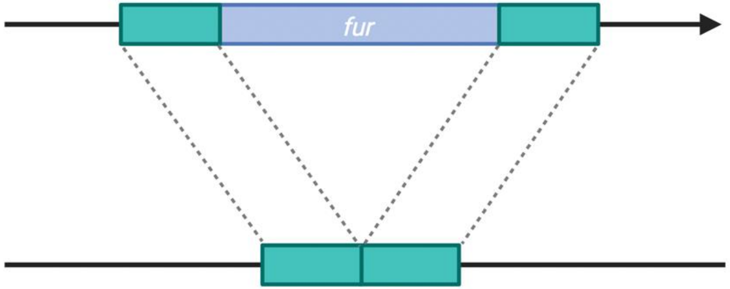
ΔtonB

- deletion of the *tonB* iron import complex
- expect low intracellular iron

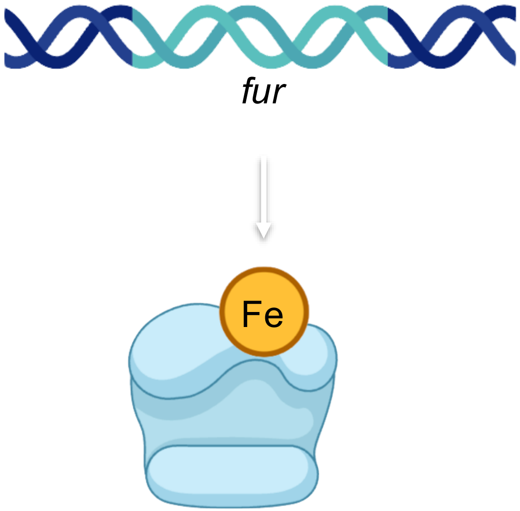


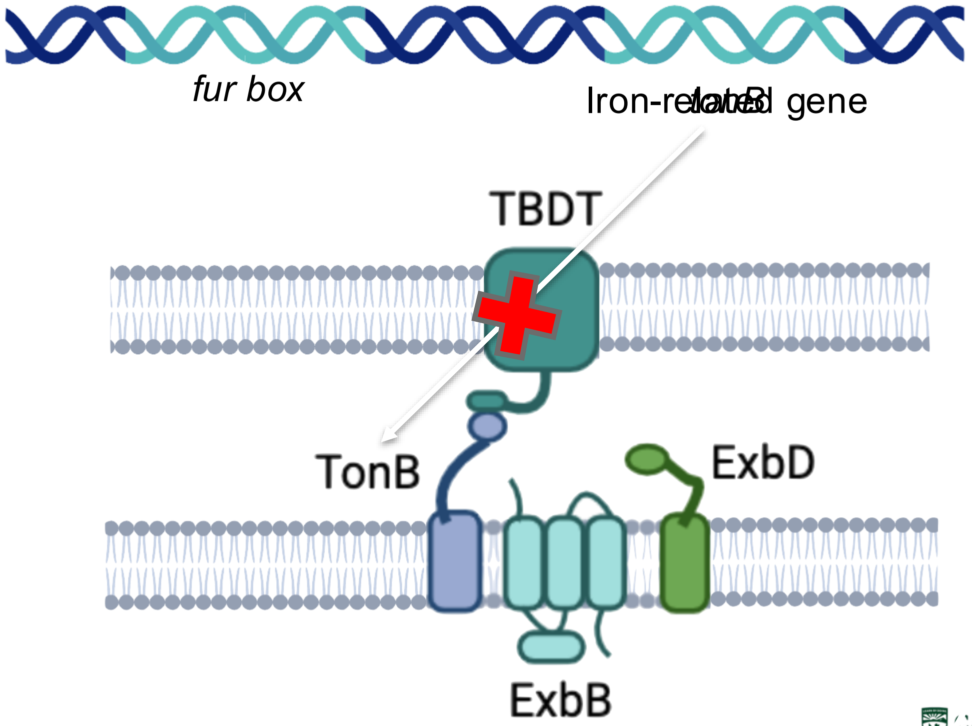
Δfur

- deletion of the *fur* iron import repressor
- expect high intracellular iron



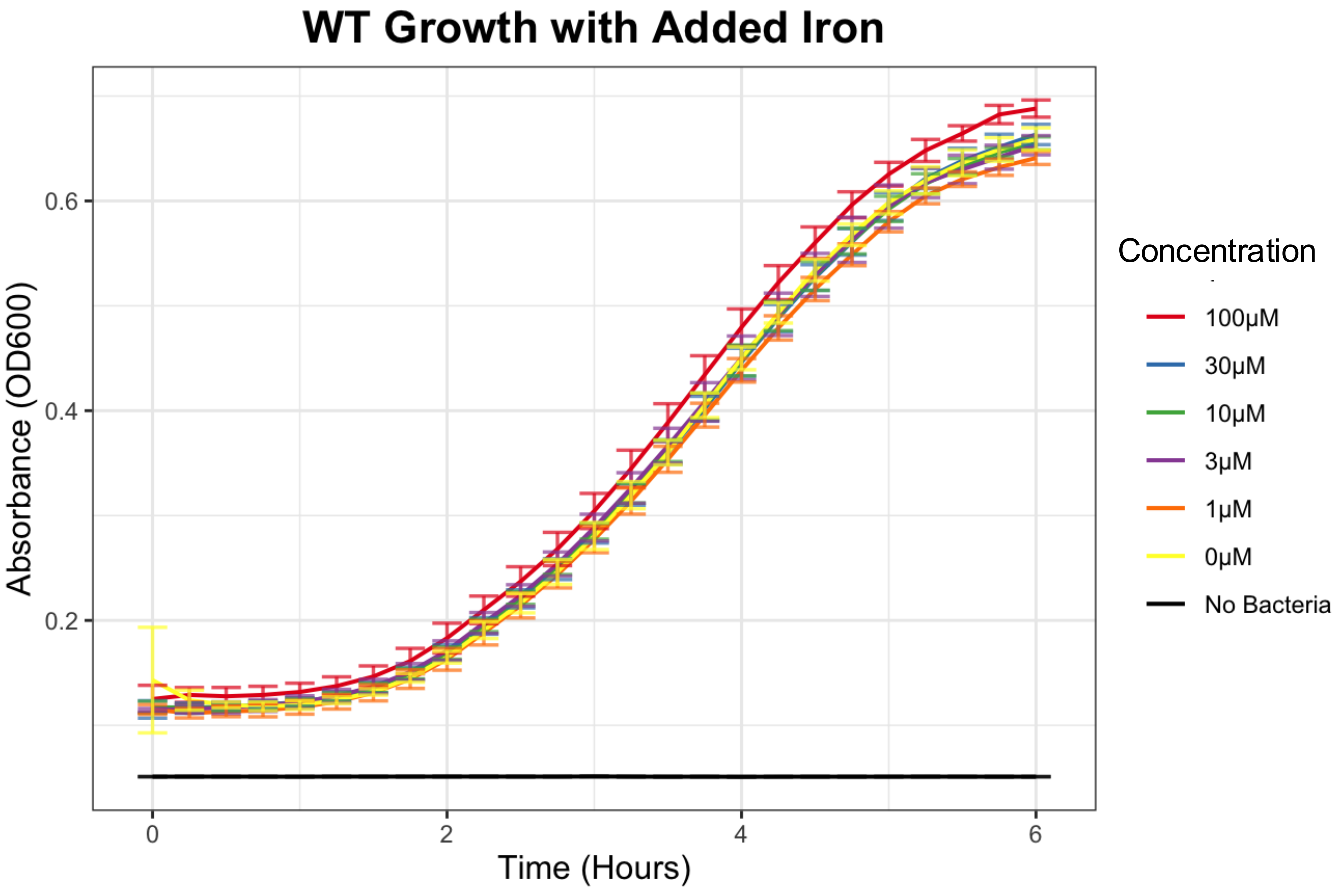
Fur and TonB are key players in iron uptake



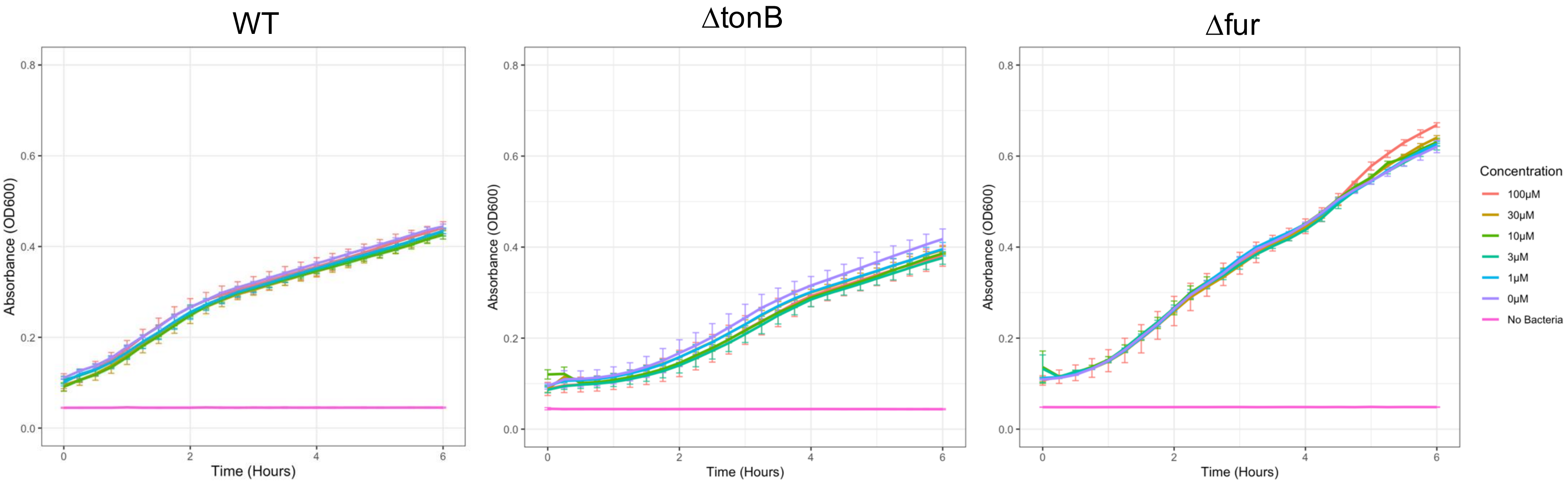


Will inhibition of iron acquisition impact these motility & attachment virulence factors?

Swimming Motility


WT Growth with Added Iron
Absorbance (OD600)
0.6
0.4
0.2
0
2
4
6
Time (Hours)
Concentration
100μM
30μM
10μM
3μM
1μM
0μM
No Bacteria

Swimming media with excess iron



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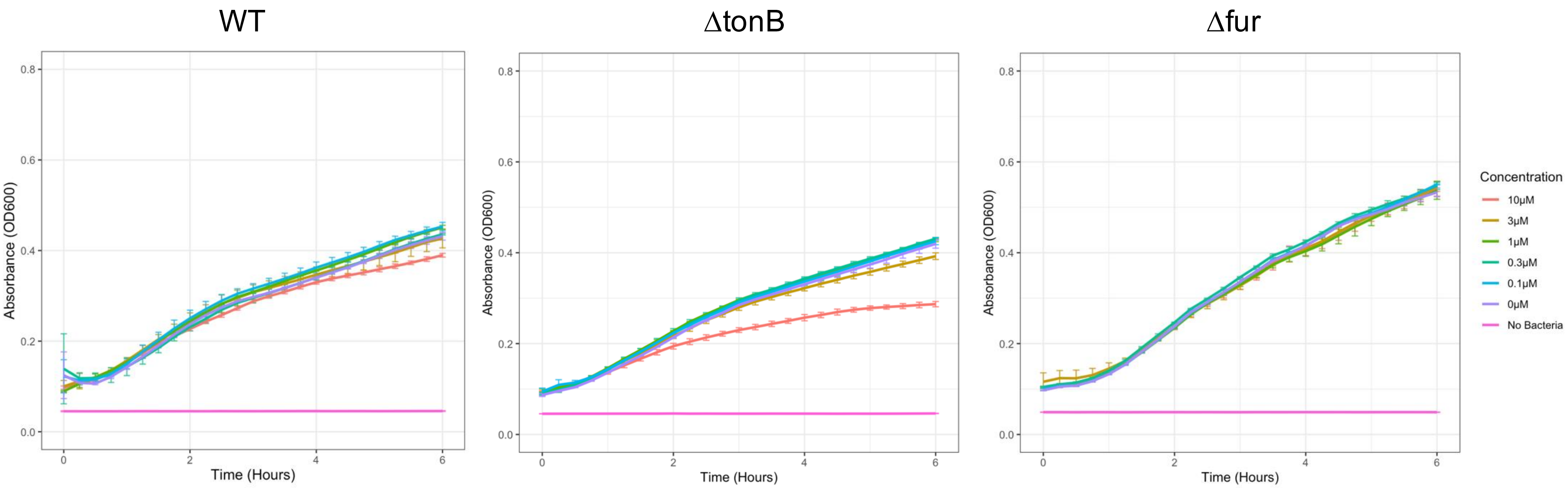
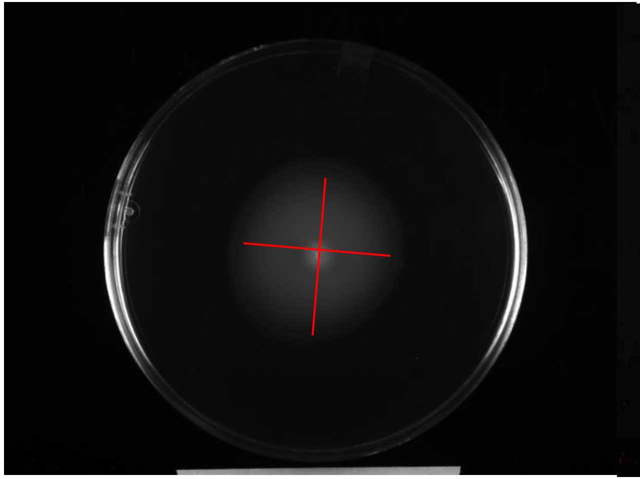
Swimming media with dipyridyl

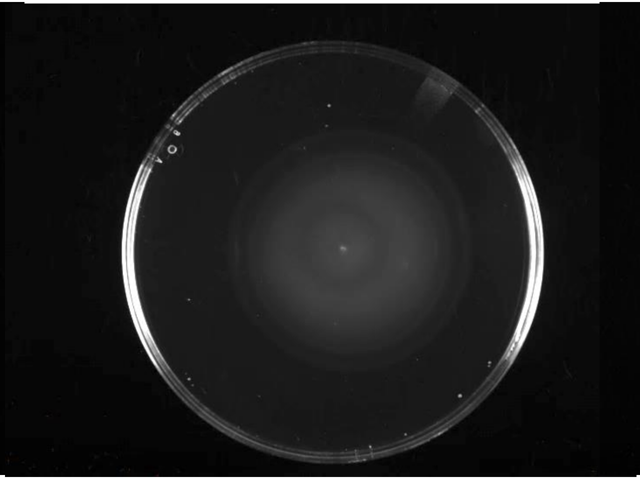


Swimming plates in control conditions

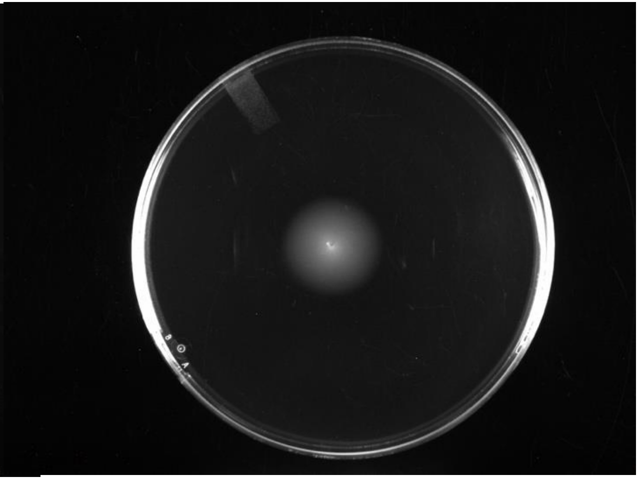
WT



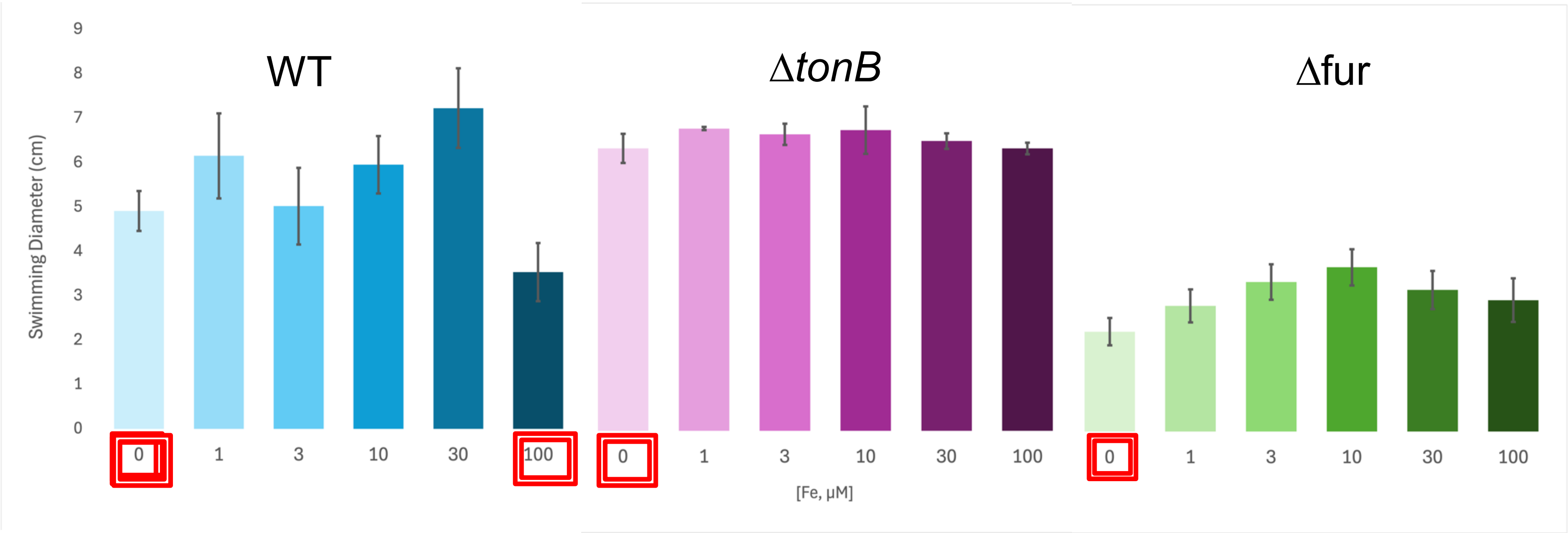
Δ*tonB*



Δ*fur*



Swimming plates with added iron

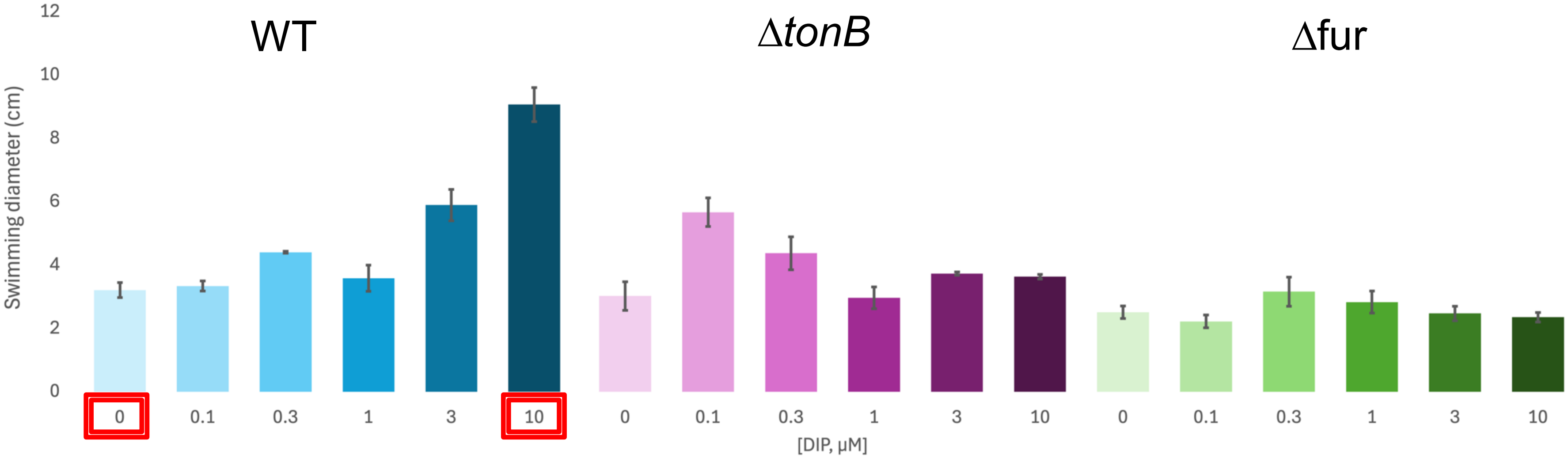


p=0.046

p=0.013

p=0.001

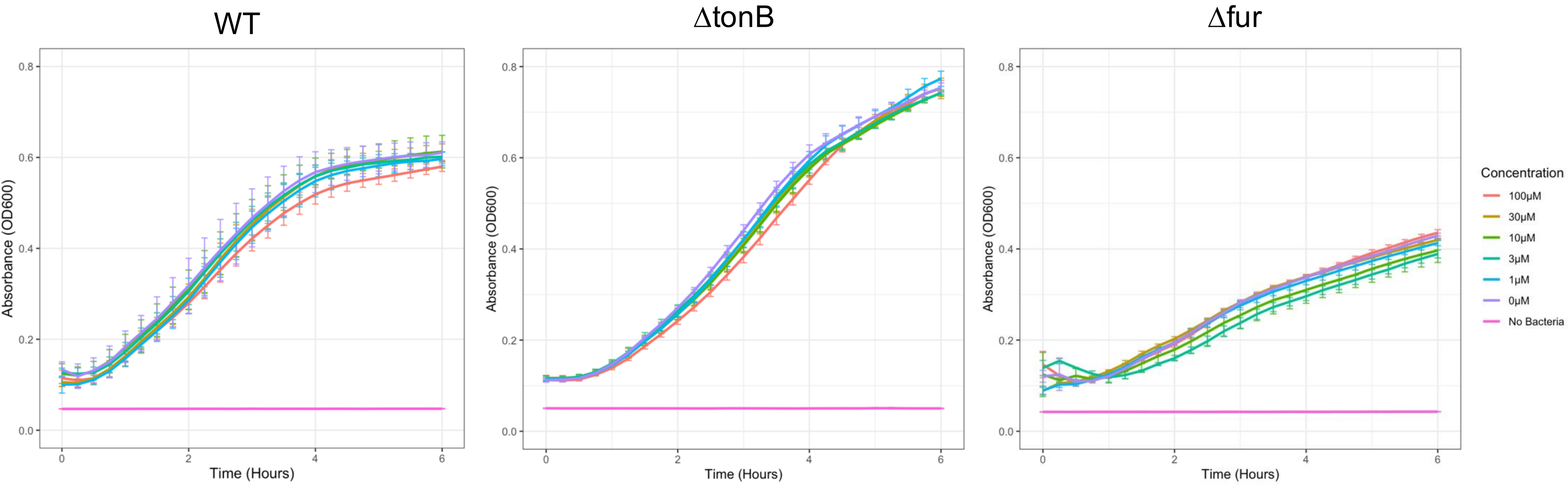
Swimming plates with chelated iron



p=<0.001

Swarming Motility

Swarming media with excess iron



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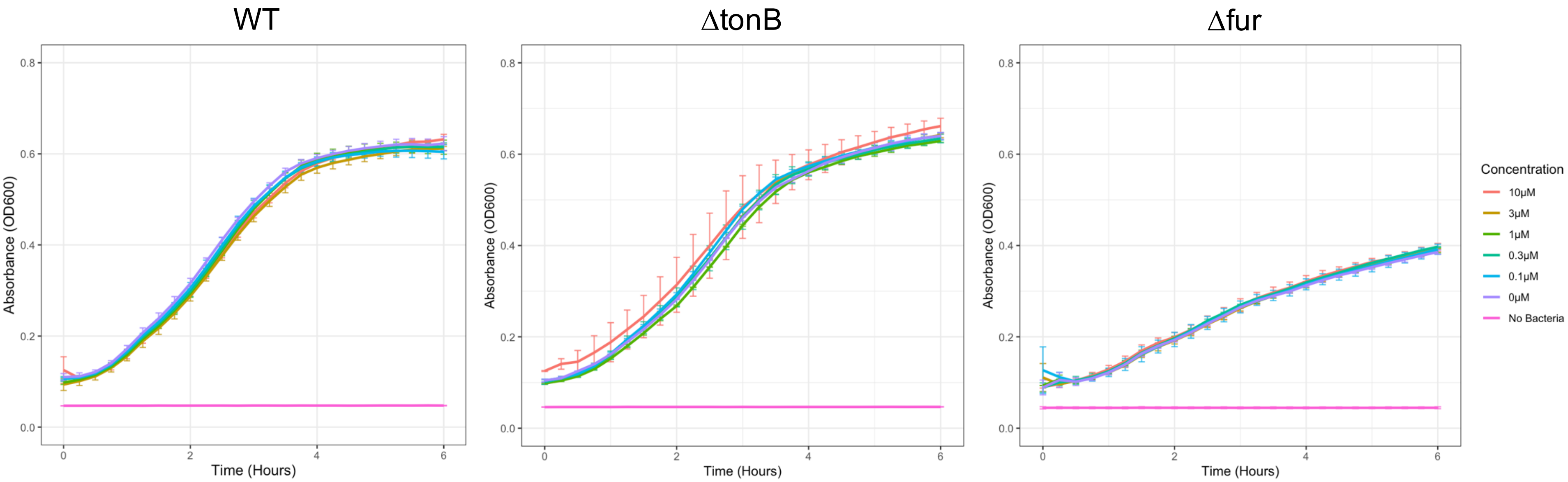
Swarming media with dipyridyl



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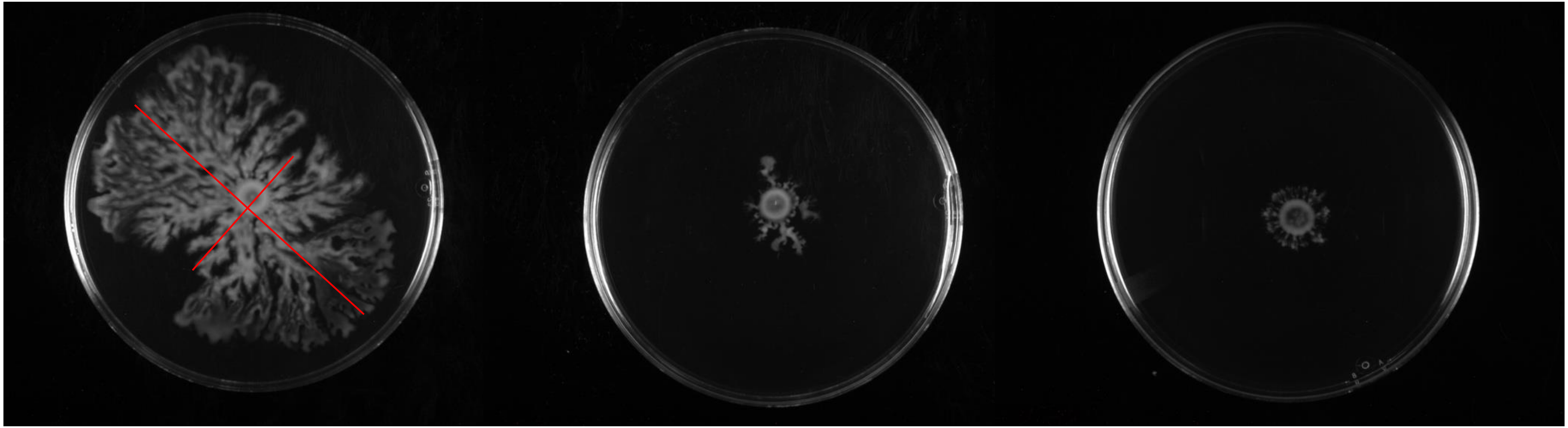
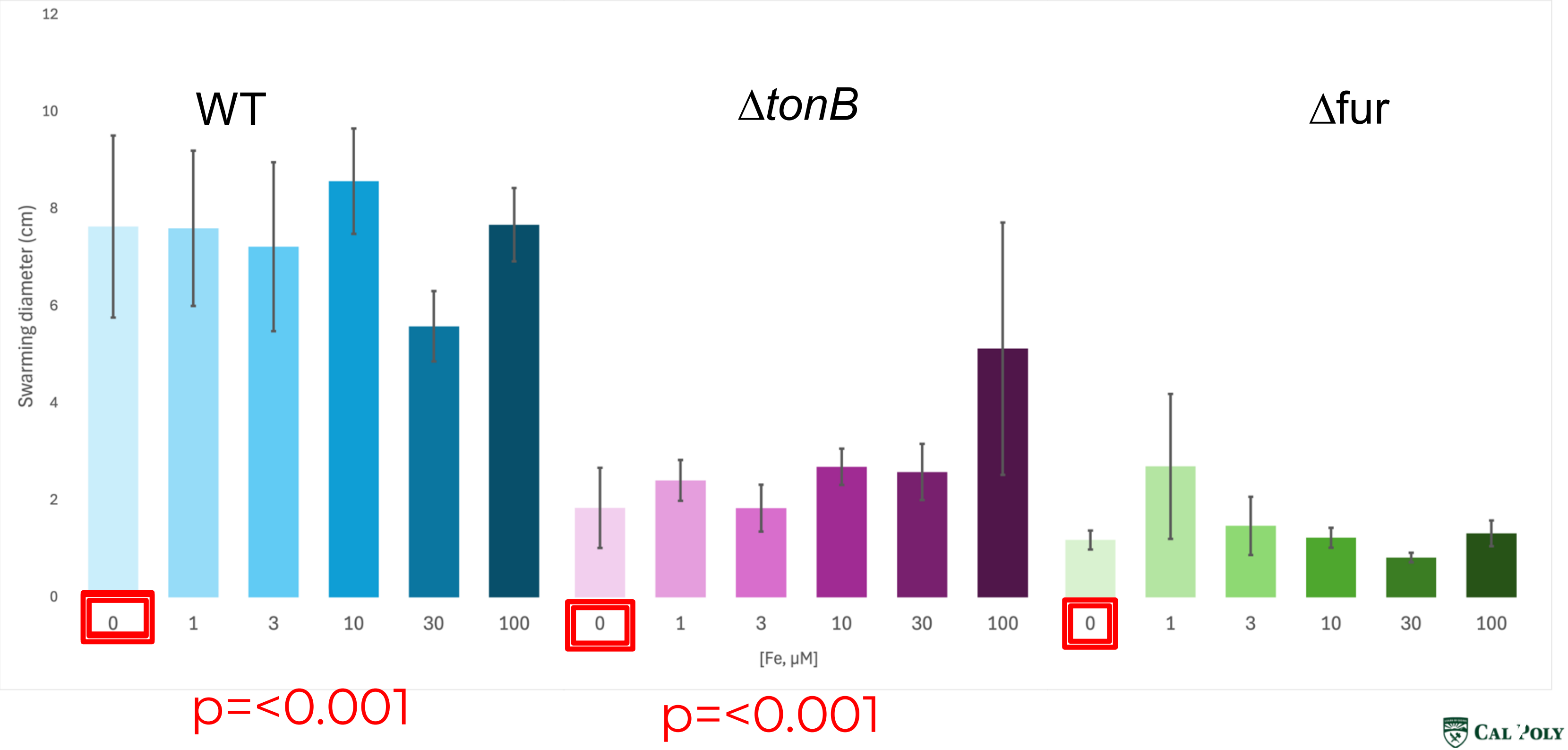
Swarming plates in control conditions

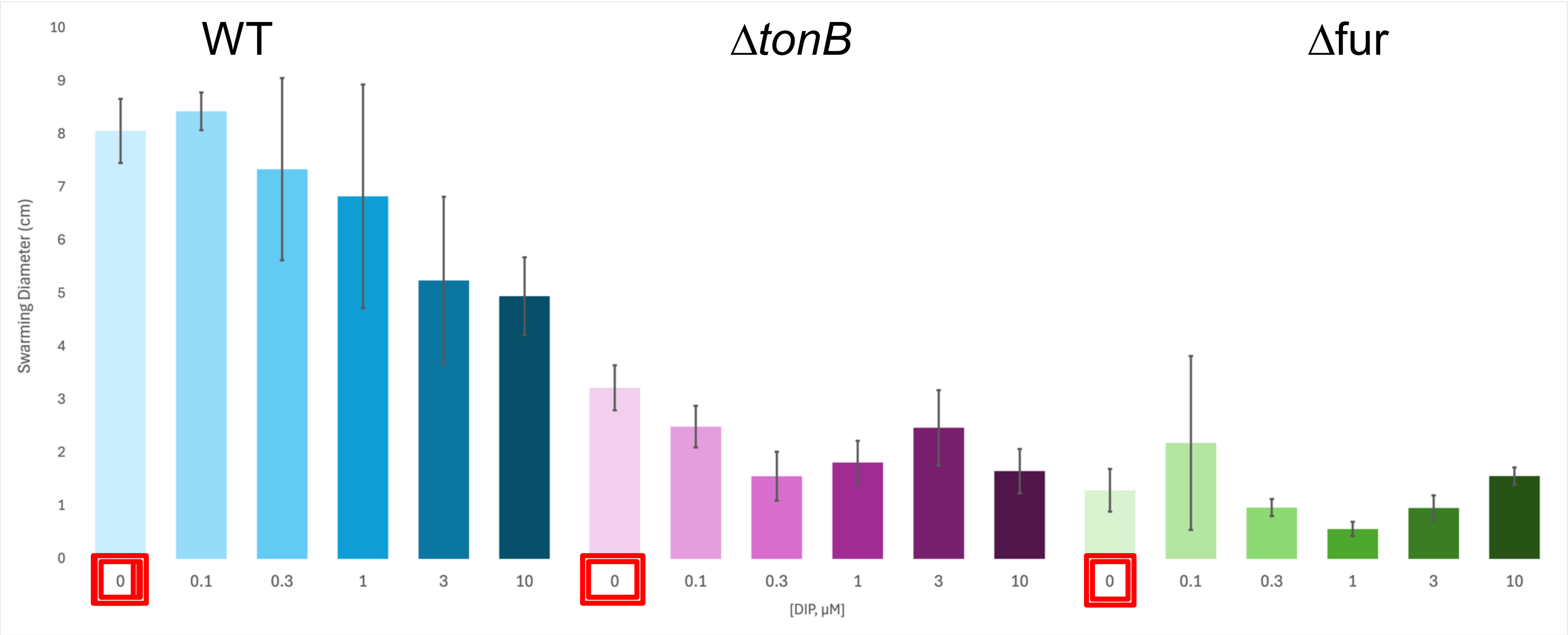
WT | *ΔtonB* | *Δfur*



Swarming plates with excess iron



Swarming plates with chelated iron



p=0.008

p=0.001

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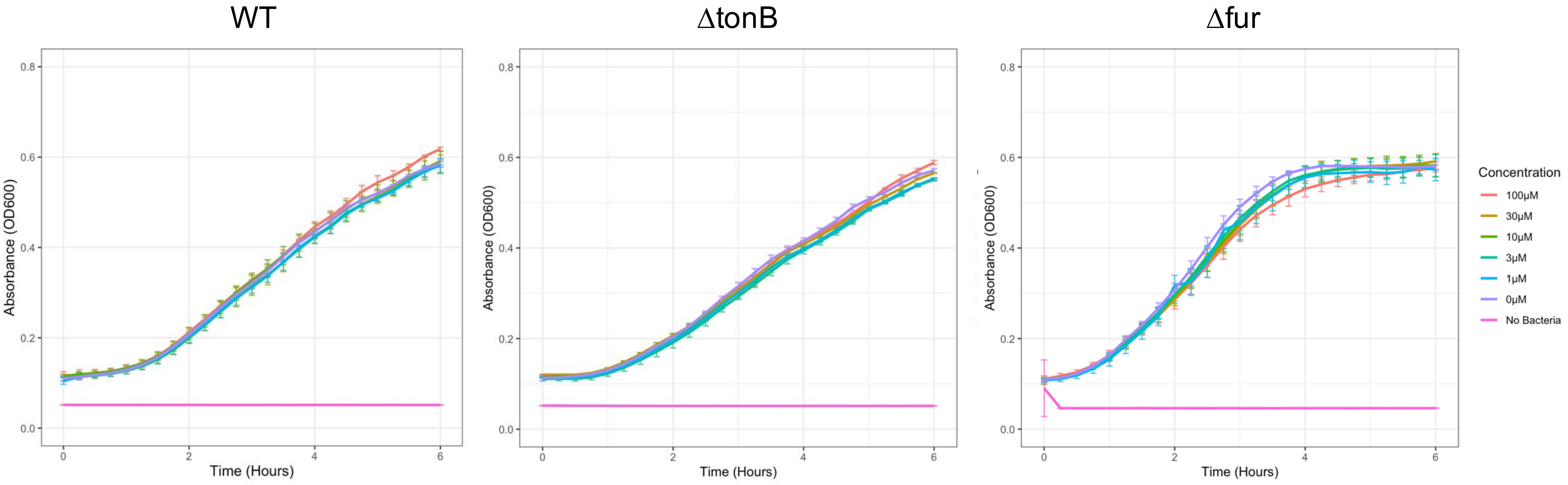
Attachment

LB with excess iron



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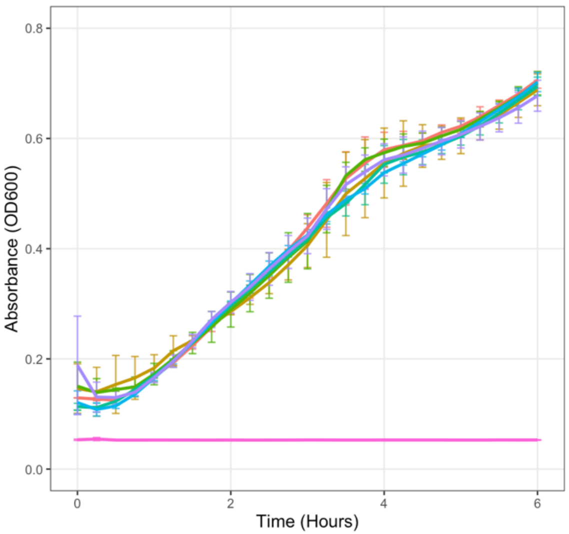
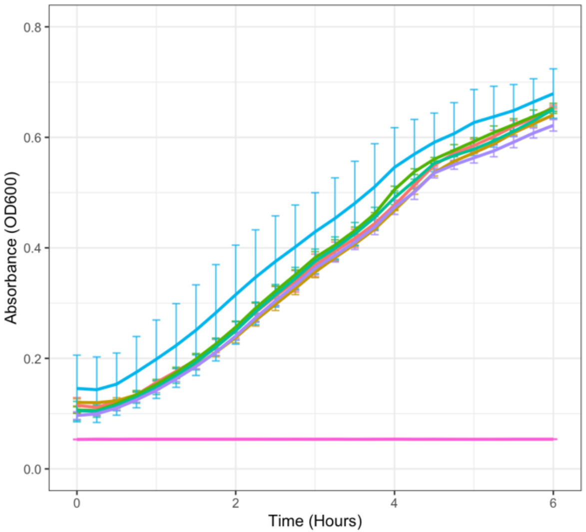
LB with dipyridyl

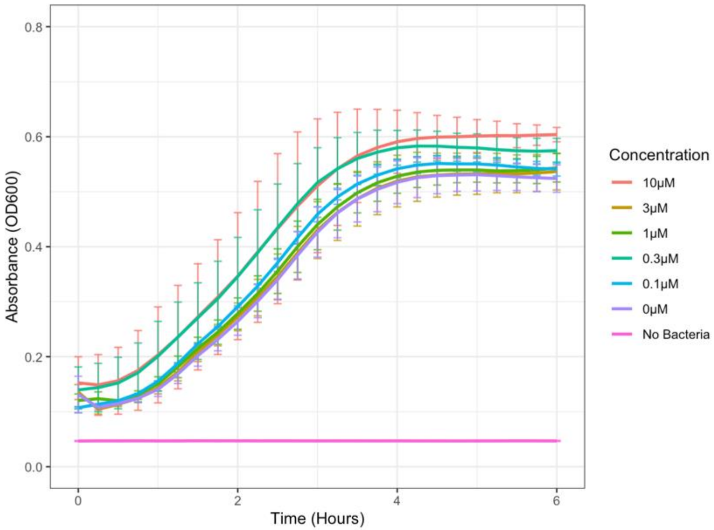
WT



ΔtonB



Δfur



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Attachment to mammalian cells

Bladder Cell Invasion Protocol

1. **Grow bladder cells to a monolayer**
2. **Wash cells in PBS**
3. **Inoculate with normalized bacteria**
4. **Incubate 1-2 hours**
5. **Wash, lyse, and plate for CFUs**

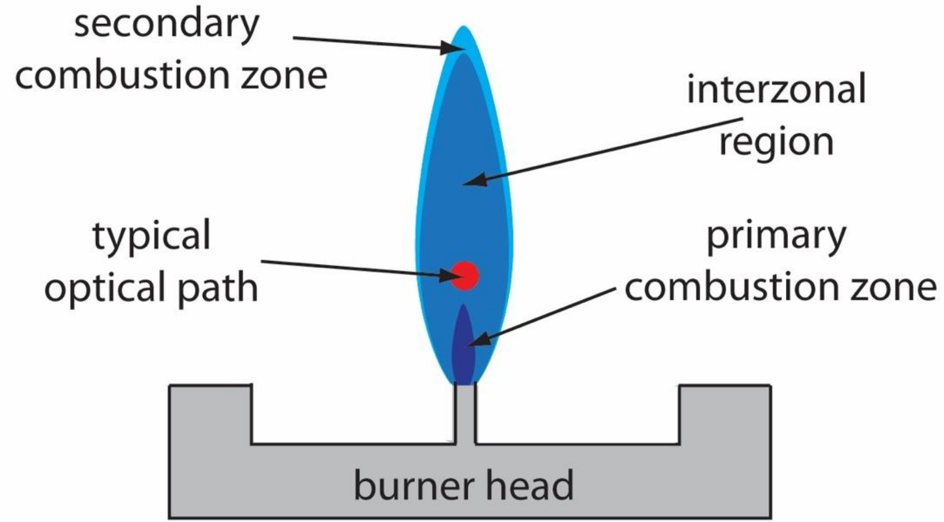
Biofilm Assay

1. **Grow UPEC in 24-well plate**
2. **Stain with crystal violet**
3. **Spec to quantify biofilm**

Done with Dr. Clement from the Biology Department with Kelsey Cliburn

Directly quantifying intracellular iron with AAS

Atomic absorption spectroscopy verifies the iron-state assumptions behind the mutants.



Culture strains

↓

Digest pellets

↓

Read Fe absorbance

↓

Calculate Fe per biomass

Done with Dr. Lehr from the Chemistry department, along with undergrads: Bohdi Leroy, Edison Deng

Automated bacterial colony counter

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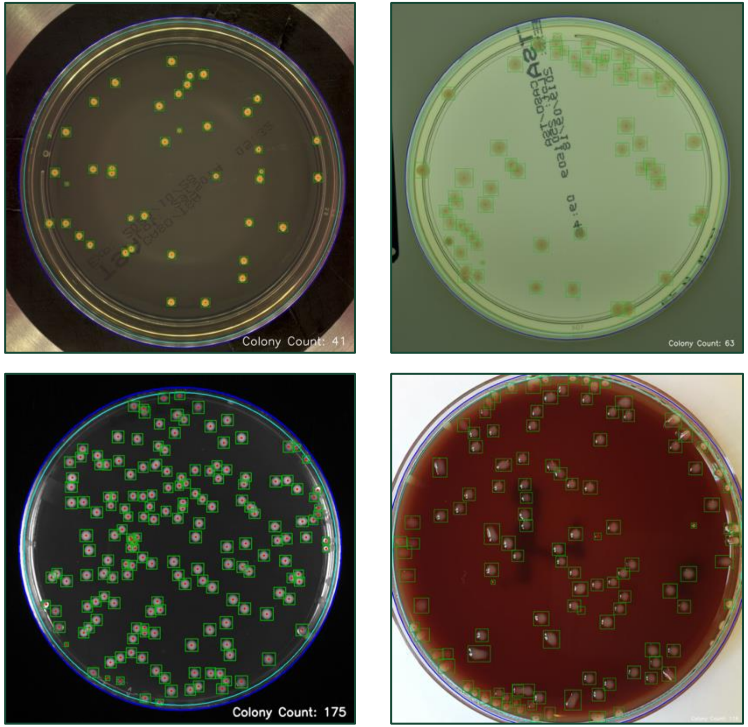
A reviewable image-analysis workflow for faster and more consistent CFU counts.

- Existing colony counters are often old or difficult to use.
- Many rely on dated image-processing algorithms rather than VLMs.
- Some tools require specialized knowledge or fail on real plate images.
- This software is designed to be easy to use and reviewable.
- Goal: accurate CFU counts with an overlay anyone can check.



Inhibition of iron acquisition systems is unlikely to increase virulence of UPEC

- Swimming may increase
- Urine is depleted of essential nutrients
- Growth limitations should counteract any increase in motility
- Next: test swimming in the presence of inhibitors

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